

AI

Section Id :	64065399794
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	9
Number of Questions to be attempted :	9
Section Marks :	27
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653230023
Question Shuffling Allowed :	No

Question Number : 62 Question Id : 6406531422431 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534756106. ✓ YES

6406534756107. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653230024
Question Shuffling Allowed :	No

Question Number : 63 Question Id : 6406531422432 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

ASK FOR PRINTED GRAPH SHEETS

Pages 1 to 8

Options :

6406534756108. ✓ Printed graph sheets were provided on time.

6406534756109. ✗ Printed graph sheets were provided late.

6406534756110. ✗ Printed graph sheets were not provided.

6406534756111. ✓ I used the graph sheets.

6406534756112. ✗ I did not use graph sheets.

Sub-Section Number :

3

Sub-Section Id :

640653230025

Question Shuffling Allowed :

No

Question Id : 6406531422433 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (64 to 68)

Question Label : Comprehension

SEARCH

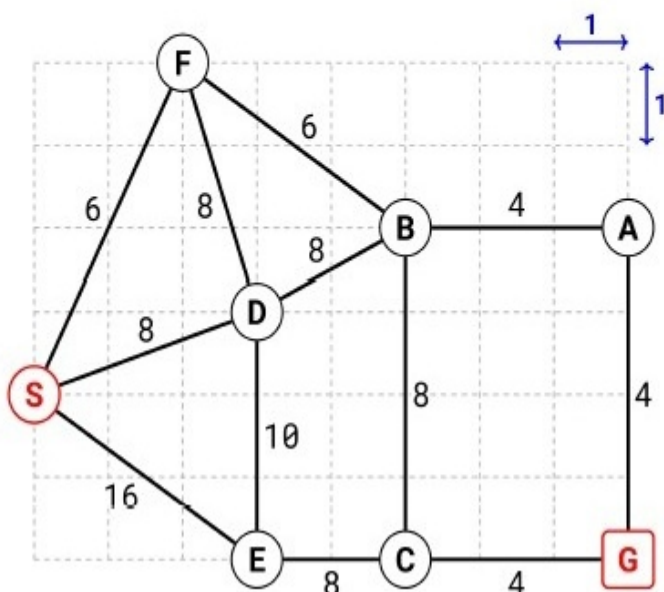
The figure shows a map on a uniform grid where each tile is 1x1 in size.

The start node is S and the goal node is G.

The MoveGen function returns nodes in alphabetical order.

Use Manhattan Distance as the heuristic function.

Tie-breaker: If several nodes have the same cost, use node labels to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 64 Question Id : 6406531422434 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by the Best First Search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,E,C,G

Question Number : 65 Question Id : 6406531422435 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by A* search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,F,B,A,G

Question Number : 66 Question Id : 6406531422436 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Branch-and-Bound search algorithm? Enter the path as a comma separated list of node labels.

Use the Branch-and-Bound variation that avoids cyclic expansions like S,E,S,E,S,E,...

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,F,B,A,G

Question Number : 67 Question Id : 6406531422437 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given map, which algorithm finds the shortest path from S to G?

Options :

6406534756116. ✖ A* Search Algorithm

6406534756117. ✖ Branch-and-Bound Search Algorithm

6406534756118. ✔ Both A* and Branch-and-Bound Search Algorithms

6406534756119. ✖ None of these

Question Number : 68 Question Id : 6406531422438 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the correct statement(s) about the given graph.

Options :

6406534756120. ✔ Heuristic is admissible.

6406534756121. ✖ Heuristic is not admissible.

6406534756122. ✖ Heuristic is admissible in some cases and not admissible in other cases.

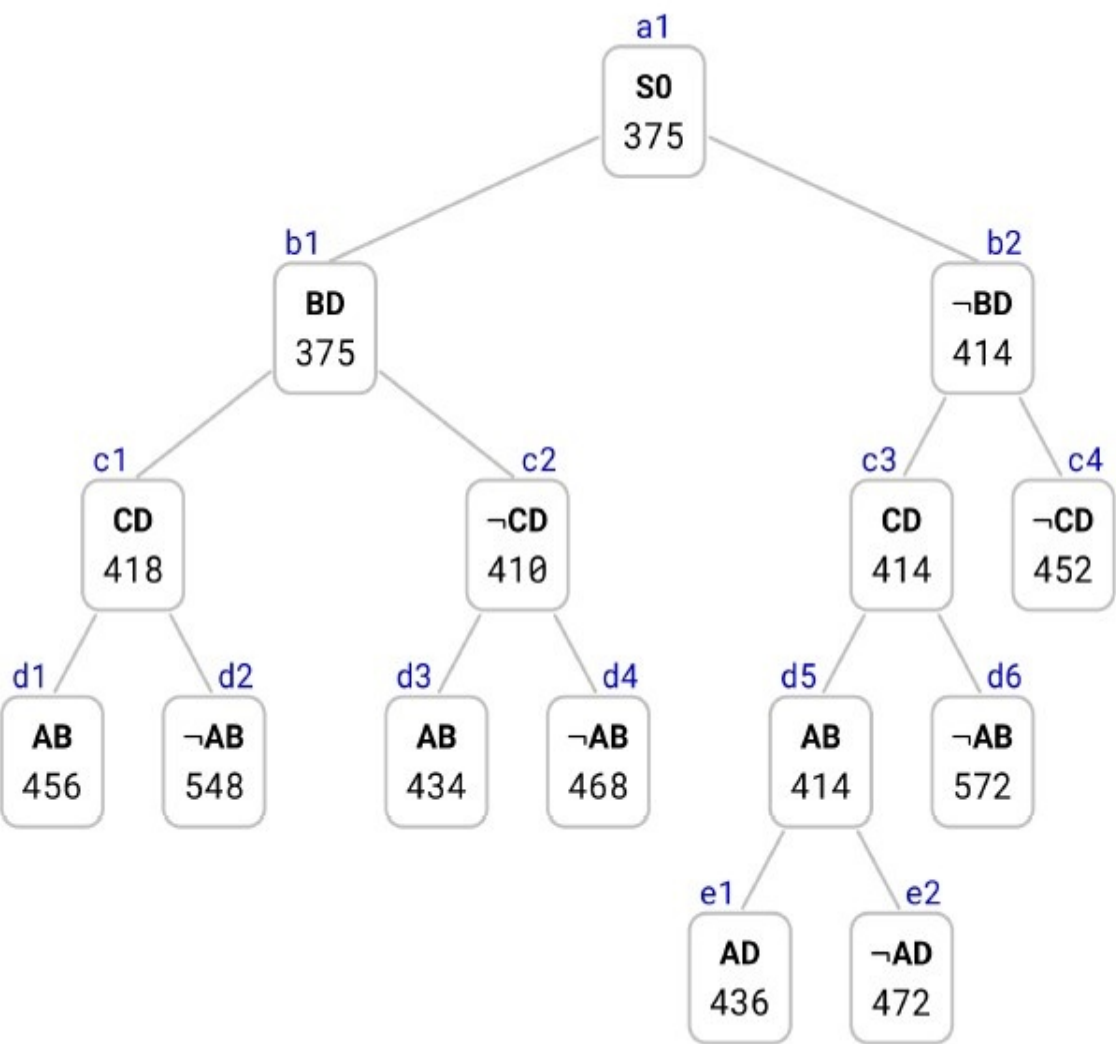
6406534756123. ✖ There is not enough information to determine admissibility.

Question Id : 6406531422439 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (69 to 73) Question Label : Comprehension TSP Branch-and-Bound

The TSP Branch-and-Bound algorithm is solving a TSP instance where the cities are A, B, C, and so on. The Branch-and-Bound search tree at the time when the algorithm has discovered the optimal tour is shown below.

Each node in the search tree displays an edge (either XY or ~XY), a cost value, and a unique reference number (a1, b1, b2, ..., c1, ..., d1, ..., e1, e2). Use the reference numbers to break ties. When required, enter the reference numbers in short answers.

What information can you glean from the search tree? Answer the sub-questions based on the information gleaned from the search tree.



Sub questions

Question Number : 69 Question Id : 6406531422440 Question Type : SA Correct Marks : 1 Question Label : Short Answer Question

Let S0 (ref. no. a1) be the first node to be refined, identify the next 4 nodes (2nd to 5th node) that are refined by the TSP Branch-and-Bound algorithm. Enter the nodes (node reference numbers) in the order they are refined.

Enter a comma separated list of node reference numbers.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: a9,b9,c9,d9

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

b1,c2,b2,c3

Question Number : 70 Question Id : 6406531422441 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Which node represents the optimal tour? Enter the node reference number in the text box, or enter NIL if it is not possible to determine the optimal tour.

Enter a node reference number.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: a9

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

d3

Question Number : 71 Question Id : 6406531422442 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the optimal tour? Enter the cost of the optimal tour in the text box, or enter NIL if it is not possible to determine the optimal tour.

Enter an integer.
NO SPACES, TABS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

434

Question Number : 72 Question Id : 6406531422443 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the number of cities in the TSP instance. Enter the number of cities in the text box, or enter NIL if it is not possible to determine the number of cities.

Enter an integer.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

5

Question Number : 73 Question Id : 6406531422444 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Start from city A, what is the path representation of the optimal tour? Enter the path representation in the text box, or enter NIL if it is not possible to determine the optimal tour.

Enter a comma separated list of cities (city labels).

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: A,B,C

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,D,E,C

A,C,E,D,B

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

4

640653230026

No

Question Id : 6406531422445 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (74 to 77)

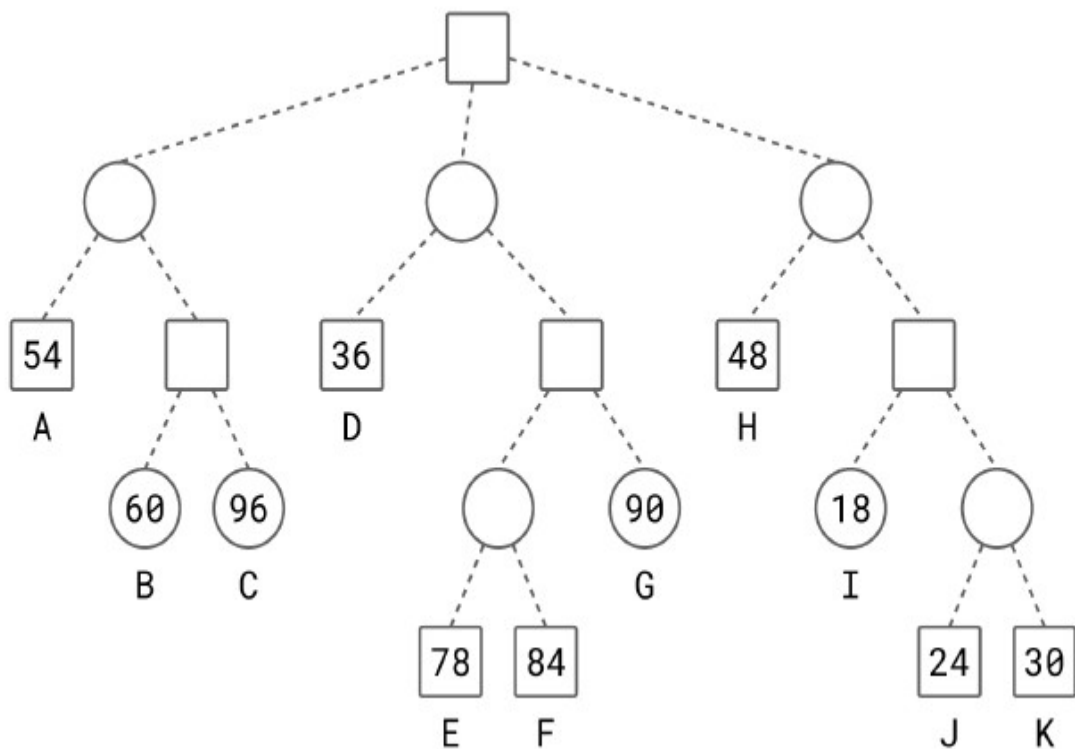
Question Label : Comprehension

GAMES

The figure shows a game tree with evaluation function values at the leaf nodes. The leaf nodes are labeled from A to K. Use these labels to enter a leaf node or a list of leaf nodes in short answers (textbox).

Tie-breaker:

When several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 74 Question Id : 6406531422446 Question Type : MSQ Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is a strategy for the MAX player?

Options :

6406534756129. ✖ D,E,F,G

6406534756130. ✖ H,I,J,K

6406534756131. ✔ D,E,F

6406534756132. ✔ H,I

Question Number : 75 Question Id : 6406531422447 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the leaf nodes in the best strategy for MAX. Enter the node labels in alphabetical order.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,C

Question Number : 76 Question Id : 6406531422448 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the leaf nodes inspected by Alpha-Beta algorithm.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,D,H

Question Number : 77 Question Id : 6406531422449 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the leaf nodes SOLVED by SSS*.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,D,H

A,D,H,B

D,H,A,B

Sub-Section Number : 5

Sub-Section Id : 640653230027

Question Shuffling Allowed : No

Question Id : 6406531422450 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (78 to 80)

Question Label : Comprehension

PROBLEM DECOMPOSITION

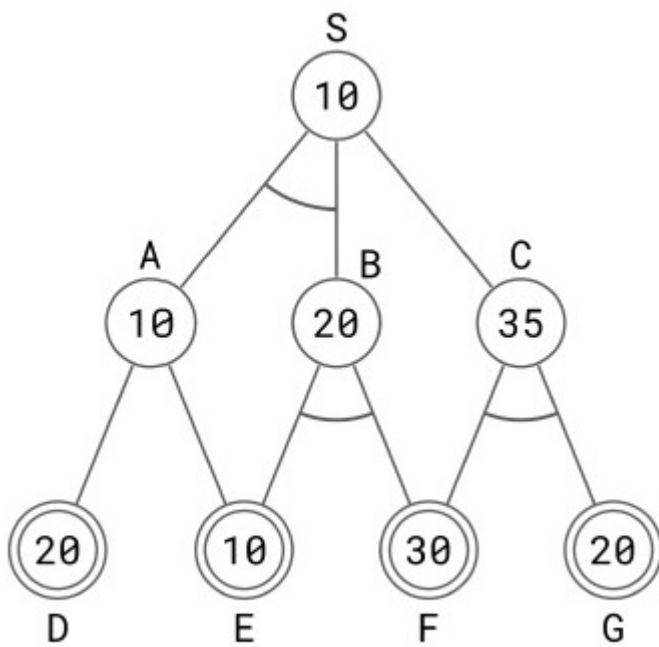
The figure shows an AND-OR graph that depicts how a problem S can be decomposed into one or more smaller problems. Nodes are uniquely identified by labels (S, A, B, ...). The number in each node is the heuristic estimate of the cost of solving that node.

Nodes shown in double lines are primitive nodes and their values are actual costs. Observe that a primitive node is added to the graph by its parent when the parent is expanded, and the primitive node is labeled as SOLVED and it will not be expanded subsequently.

The cost of each edge is 10 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.

Tie-breaker 2: For AND nodes, expand the unsolved branch with the highest cost



Use AO* algorithm to solve S, then answer the subquestions.

Sub questions

Question Number : 78 Question Id : 6406531422451 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the nodes (including S) expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,C,B

C,B

Question Number : 79 Question Id : 6406531422452 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the value of the start node S after each node is expanded. Enter the value of S after each node is expanded.

Enter a comma separated list of numbers.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

45,50,80

50,80

Question Number : 80 Question Id : 6406531422453 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the final value of the start node S computed by AO*?

Enter a number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

Question Id : 6406531422454 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

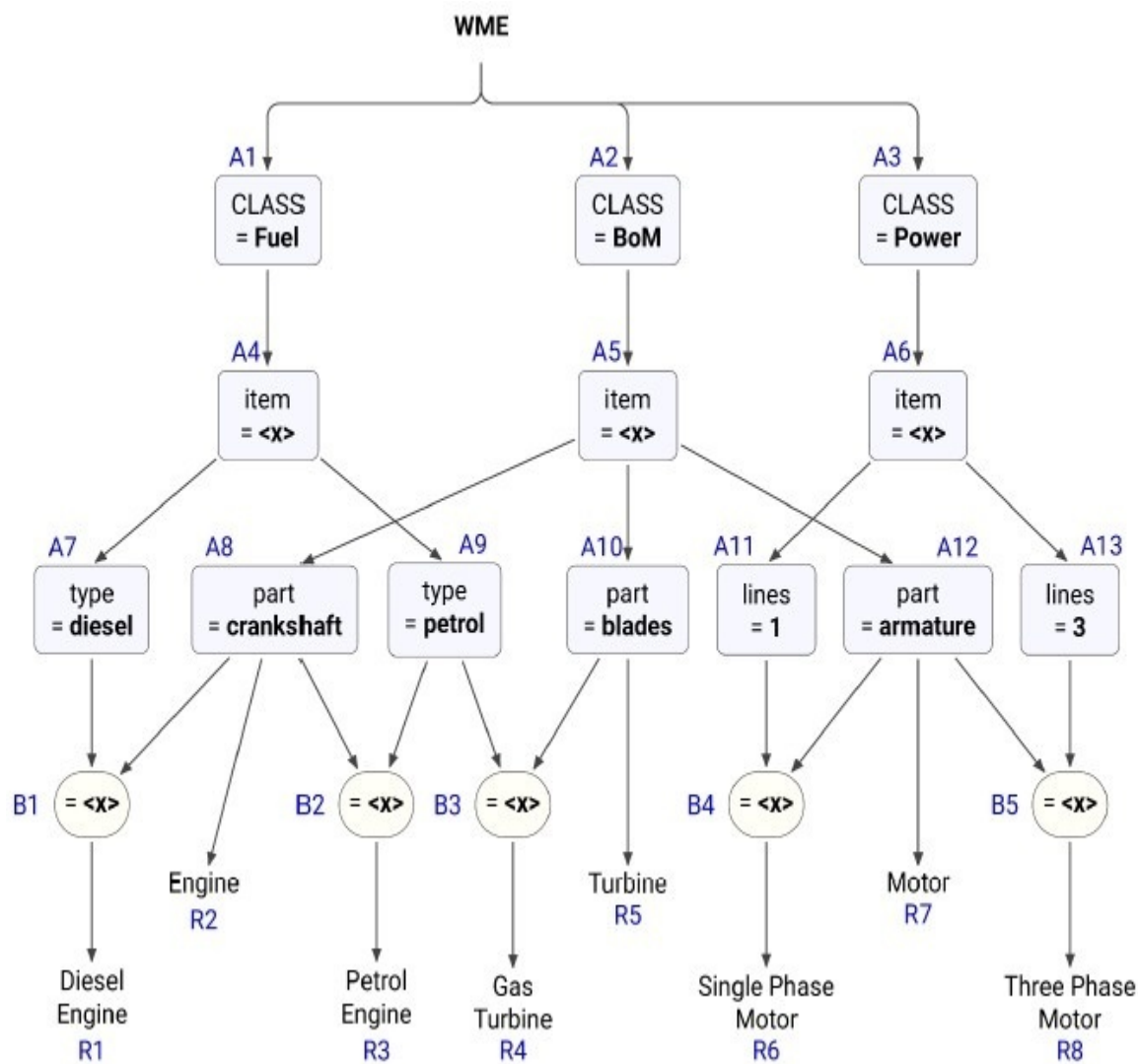
Question Numbers : (81 to 83)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A Rete Net for a rule based system to identify engines, turbines and motors is shown in the figure. BoM refers to Bill-of-Materials, which is a list of parts and their quantities needed for assembling a machine.

The nodes in the network are uniquely identified by labels: A1,A2,... for Alpha nodes; B1,B2,... for beta nodes; R1,R2,... for rules.



The Working Memory contains the following WMEs uniquely identified by timestamps (sequence numbers). Assume that WMEs reside in the appropriate Alpha node, and Beta nodes simply point to elements in the ancestor Alpha nodes.

```

101. (Fuel ^item M59 ^type diesel)
102. (Fuel ^item M26 ^type petrol)
103. (Fuel ^item M37)
104. (BoM ^item M48 ^part armature)
105. (Power ^item M48 ^lines 1)
106. (BoM ^item M37 ^part armature)
107. (BoM ^item M15 ^part crankshaft)

```

The sub-questions are based on the first Match-Resolve-Execute cycle. Determine the locations of the WMEs and compute the conflict set, then answer the sub-questions.

Sub questions

Question Number : 81 Question Id : 6406531422455 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples occur in the conflict set?

Options :

- 6406534756139. ✖ (Diesel-Engine, 101, 107)
- 6406534756140. ✖ (Petrol-Engine, 107, 102)
- 6406534756141. ✔ (Engine, 107)
- 6406534756142. ✔ (Single-Phase-Motor, 105, 104)
- 6406534756143. ✔ (Motor, 104)
- 6406534756144. ✔ (Motor, 106)

Question Number : 82 Question Id : 6406531422456 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Specificity** for conflict resolution then which rule-data tuple will fire in the first round?

Options :

- 6406534756145. ✖ (Diesel-Engine, 101, 107)
- 6406534756146. ✖ (Petrol-Engine, 107, 102)
- 6406534756147. ✖ (Engine, 107)
- 6406534756148. ✔ (Single-Phase-Motor, 105, 104)
- 6406534756149. ✖ (Motor, 104)
- 6406534756150. ✖ (Motor, 106)

Question Number : 83 Question Id : 6406531422457 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** for conflict resolution which rule-data tuple will fire in the first round?

Options :

- 6406534756151. ✖ (Diesel-Engine, 101, 107)
- 6406534756152. ✖ (Petrol-Engine, 107, 102)
- 6406534756153. ✔ (Engine, 107)
- 6406534756154. ✖ (Single-Phase-Motor, 105, 104)
- 6406534756155. ✖ (Motor, 104)
- 6406534756156. ✖ (Motor, 106)

Sub-Section Number :

6

Sub-Section Id :

640653230028

Question Shuffling Allowed :

No

Question Id : 6406531422458 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (84 to 87)

Question Label : Comprehension

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

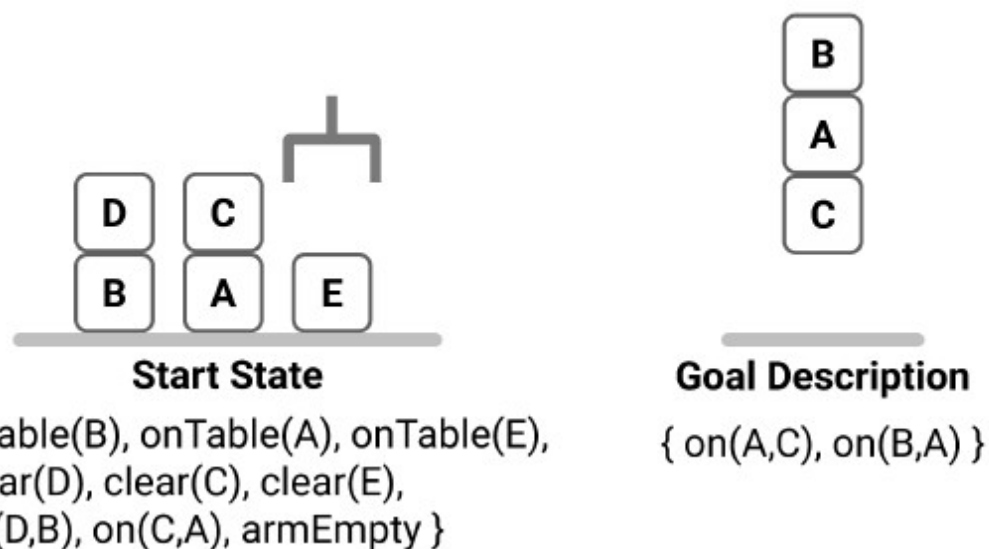
Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

Consider the following planning problem.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 84 Question Id : 6406531422459 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions in the start state?

Options :

- 6406534756157. ✖ Putdown(D)
- 6406534756158. ✔ Unstack(D, B)
- 6406534756159. ✔ Unstack(C, A)
- 6406534756160. ✔ Pickup(E)
- 6406534756161. ✖ Pickup(A)
- 6406534756162. ✖ Stack(B, A)
- 6406534756163. ✖ Stack(A, C)

Question Number : 85 Question Id : 6406531422460 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions in the goal state?

Options :

- 6406534756164. ✖ Putdown(D)
- 6406534756165. ✖ Unstack(D, B)
- 6406534756166. ✖ Unstack(C, A)
- 6406534756167. ✖ Pickup(E)
- 6406534756168. ✖ Pickup(A)

6406534756169. ✓ Stack(B, A)

6406534756170. ✓ Stack(A , C)

Question Number : 86 Question Id : 6406531422461 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex action pairs in Layer 1?

Options :

6406534756171. ✓ Unstack(D, B), Pickup(E)

6406534756172. ✓ Unstack(D, B), Unstack(C, A)

6406534756173. ✗ Pickup(E), Stack(B, A)

6406534756174. ✗ Pickup(E), Stack(A, C)

Question Number : 87 Question Id : 6406531422462 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex proposition pairs in Layer 1?

Options :

6406534756175. ✓ clear(B), holding(C)

6406534756176. ✓ clear(B), clear(A)

6406534756177. ✗ clear(E), on(D, B)

6406534756178. ✗ clear(E), holding(D)

Sub-Section Number :

7

Sub-Section Id :

640653230029

Question Shuffling Allowed :

No

Question Id : 6406531422463 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (88 to 90)

Question Label : Comprehension

CONSTRAINT SATISFACTION

Consider a map colouring problem with 3 regions {A,B,C}, where all three regions are adjacent to each other, and the regions are being processed in the order A,B,C.

The domains are:

$$D_A = \{r, b, g\}$$

$$D_B = \{r, b\}$$

$$D_C = \{r, b\}$$

Draw the constraint graph for the above problem and answer the sub-questions.

Sub questions

Question Number : 88 Question Id : 6406531422464 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP arc consistent?

Options :

6406534756179. ✓ Yes

6406534756180. ✗ No

6406534756181. ✗ Cannot be determined

Question Number : 89 Question Id : 6406531422465 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP path consistent?

Options :

6406534756182. ✓ No

6406534756183. ✗ Yes

6406534756184. ✗ Cannot be determined

Question Number : 90 Question Id : 6406531422466 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Does the given CSP have a solution? Enter the solution for regions A,B,C as a comma separated list of colours. Enter NIL if there is no solution.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: r,b,r

Response Type : Alphanumeric

Evaluation Required For SA : Yes